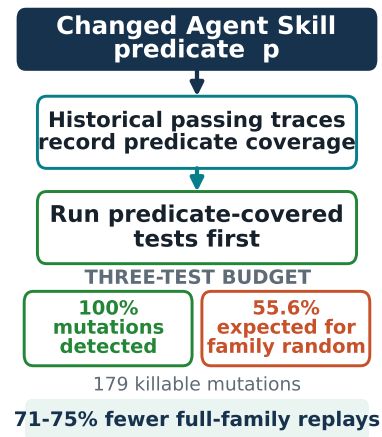


Graphical Table of Contents

Title: From Skill-Level to Predicate-Level: Budgeted Regression Test Selection for Evolving LLM Agent Skills

Authors: Liang Song* and Zhai Jiabao



PredicateImpact records predicate-level coverage from passing agent-skill traces and prioritizes tests that exercised the changed predicate. Across two disjoint ALFWorld and ScienceWorld confirmations, it detected all 179 killable mutations within a three-test budget, compared with about 56% expected detection for skill-family random selection, while avoiding 71–75% of full-family test executions.